

## Ordered pairs

An ordered pair gives horizontal and vertical coordinates.

$(x, y)$

**Why does reversing the coordinates usually change the point?**

## Plot a point

$(-4, 3)$

**Where is this point, and which quadrant contains it?**

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**A point on an axis**

$(0, -5)$

**Where should we plot this point?**

## Coordinate scales

On a graph, one horizontal grid step represents 2 units; one vertical step represents 5 units.

**What are the coordinates three steps right and two steps down from the origin?**

## A graph of an equation

A graph consists of all ordered pairs that satisfy its equation.

**How could we decide whether a proposed point belongs to a graph?**

## Test a point

$$y = 3x - 2, \quad (2, 4)$$

**Does this point satisfy the equation?**

## Build a linear graph

$$y = -2x + 4$$

**What graph results from plotting points for  $x = -1, 0, 1,$  and  $2$ ?**

## Build a curved graph

$$y = x^2 - 1$$

**What shape appears when we plot points for  $x = -2, -1, 0, 1,$  and  $2$ ?**

## Two intercepts

An x-intercept has  $y = 0$ . A y-intercept has  $x = 0$ .

**Why are these the coordinates we set equal to zero?**

## Find intercepts

$$2x + 3y = 12$$

**What are the intercepts, written as ordered pairs?**

## More than one x-intercept

$$y = x^2 - 9$$

**Where does this graph meet the x-axis?**

## An intercept at the origin

$$y = x^2$$

**Which intercepts does this graph have?**

## Choose a viewing window

$$y = 2x - 30$$

**What calculator window would show both intercepts clearly?**

## A missing graph feature

A calculator window shows only  $-10 \leq x \leq 10$  and  $-10 \leq y \leq 10$ .

$$y = x^2 - 36$$

**Could this view hide an important feature of the graph?**

## Intersections

An intersection point satisfies the equations of both graphs.

$$y = 2x + 1, \quad y = 7$$

**What ordered pair is their intersection?**

## Distance between points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**How does a right triangle lead to this distance formula?**

## Calculate a distance

$$A = (-3, 2), \quad B = (3, 10)$$

**What is the straight-line distance from A to B?**

## Distance and route length

Travel from  $(0, 0)$  to  $(6, 8)$ , moving only horizontally and vertically. Each unit is one kilometer.

**How does the shortest allowed route compare with the straight-line distance?**

## Midpoint

$$M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

**Why do we average the two x-coordinates and the two y-coordinates?**

## Calculate a midpoint

$$A = (-5, 6), \quad B = (7, -2)$$

**What point lies halfway between A and B?**

## Recover an endpoint

A segment has endpoint  $A = (-2, 5)$  and midpoint  $M = (3, 1)$ .

**What is its other endpoint?**